TODO: Paper Title

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Abstract

Background. TODO: 1–2 sentences on the problem and why it matters.
Aims. TODO: the research questions or specific objectives.
Method. TODO: dataset, models, approaches compared.
Results. TODO: headline numbers, ordered by RQ.
Conclusions. TODO: what this means and the takeaway for practitioners.

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Keywords and phrases issue classification, retrieval-augmented generation, fine-tuning, large language models, mining software repositories

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1 Introduction

2 Related Work

Early IRC studies employed data mining techniques and ML classifiers [1, 18, 12]. For example, Antoniol et al. [1] leveraged a naive Bayes classifier to distinguish bugs from other issue reports. However, these methods showed low performance due to their limited semantic understanding of the issue report text [13].

To overcome this, recent methods have adopted transformer models due to their strong contextual understanding [6, 4, 24, 16, 17]. These studies typically fine-tune the models on historical issues in a supervised setting. For instance, Izadi et al. [16] fine-tuned a RoBERTa [21] model to classify issues into *Bugs*, *Features*, or *Questions*. Similarly, Trautsch et al. [24] trained a seBERT [25] model to classify issues into the same three categories. Although transformer-based approaches achieve strong results, they rely on large amounts of project-specific data to be effective, hindering their generalizability on projects with a small number of labeled issues [13, 16]. Additionally, they struggle to accurately classify minority classes with fewer training examples [17, 6].

These challenges have motivated the exploration of LLMs, which exhibit strong text classification capabilities thanks to extensive pretraining on massive corpora. [13, 3, 2, 5, 26, 8]. Studies show that LLMs perform well in zero-shot settings [8, 2]. SOTA LLM-based methods achieved the best performance by fine-tuning the models on labeled datasets [13, 3]. For example, Aracena et al. [3] instruction-tuned several models on the NLBSE '23 and '24 datasets.

3 Approach

3.1 Problem Formulation

Given a query issue report, we classify it into only one of the three labels: *bug*, *feature*, or *question*. Only the issue's title and description text are used for classification with no other signals from the issue tracker. The defined label space is adopted directly from prior IRC studies [2, 3, 16, 18, 1].

3.2 Indexing and Retrieval

To prepare the issues for retrieval based on textual similarity, we first embed their textual content (i.e., *title* + *description*). We do not embed the issue labels. Instead, we retain them as metadata associated with each embedded issue. To generate the embeddings, we use the `all-MiniLM-L6-v2` model from the Sentence-Transformers library [22]. We selected this model due to its lightweight architecture, fast performance, and effectiveness on issue report text [15]. The embeddings are then indexed using the Faiss vector database library [10], which provides efficient high-dimensional similarity retrieval. At the retrieval stage, the query issue is first embedded with the same model. Next, Faiss calculates the cosine similarity of indexed issue reports to the query, and returns the top-K nearest neighbors, ordered by descending similarity.

3.3 RAGTAG

RAGTAG is our proposed retrieval-augmented few-shot prompting approach for IRC. RAGTAG uses the retrieval pipeline described in Section 3.2 to retrieve the top-K nearest neighbors to the query issue, and present them as classified examples to the LLM in a few-shot prompt. This enables a more accurate classification, based on in-context learning [5]. The choice to present the top-K nearest neighbors as examples is based on two insights from prior research: First, retrieval-based example selection outperforms random selection [19]. Second, the most similar neighbors tend to share the same labels [9]; therefore, they are informative for the classification.

We create the few-shot prompt in three parts. First, the system message frames the classification task and the label space. Second, the top-K nearest neighbors are listed in retrieval order (descending similarity), each formatted as *title + description + label*. Finally, the query issue is appended to the end of the prompt, ensuring it receives high attention from the LLM [20]. $K = 0$ produces a zero-shot prompting baseline, similar to prior work [7].

We evaluate RAGTAG at top-K values in $\{0, 1, 3, 6, 9\}$. Details for the K value grid selection are provided in Section 5.1. Since the median prompt size at $K = 9$ in our dataset is ∼6,300 tokens, the maximum sequence length is set to 8,192 tokens. Roughly 30% of the context quota is reserved for the query issue and 70% for the neighbors. If the neighbors exceed their overall quota, they are truncated proportionally based on their length (longer neighbors get more truncation). The query issue gets truncated if it exceeds its budget as well. Truncation happens on the right side of the description text, preserving all titles and labels (if any).

The LLM is instructed to generate the predicted label wrapped in XML tags (e.g. `<label>bug</label>`). The labels of the presented examples also follow the same format. This technique enables easy detection of the predicted label from the LLM's generation, and reduces variance in LLM outputs caused by prompt format sensitivity [23]. We use a regular expression to extract the predicted label. Unparseable outputs are marked as *invalid*.

3.4 VOTAG

VOTAG is our proposed non-parametric retrieval IRC method. VOTAG uses the retrieval pipeline described in Section 3.2 to retrieve the top-K nearest neighbors of the query issue report, then predicts the final label with a similarity-weighted vote on their labels. This approach follows the standard K-nearest-neighbor classification setup [9], with neighbors

weighted by their similarity to the query [11].[1] Formally, for a query issue q with retrieved neighbors $\{(n_i, l_i, s_i)\}_{i=1}^{K}$, where n_i is the i-th nearest neighbor with label l_i and cosine similarity s_i to q, the label is predicted by:

$$\hat{y} = \arg\max_{c \in \mathcal{L}} \sum_{i=1}^{K} s_i \cdot \mathbf{1}[l_i = c], \tag{1}$$

where $\mathcal{L}$ is the label set and $\mathbf{1}[\cdot]$ is the indicator function. VOTAG breaks ties by returning the label of the most similar neighbor in the tied classes.

VOTAG has three main roles: First, it serves as a fast and cost-efficient baseline that LLM-based approaches must clear to justify their computational cost. Second, it is used as an ablation to distinguish the contributions of the LLM and retrieval in the RAGTAG method. Third, VOTAG's performance as a function of K informs the K value grid evaluated for RAGTAG.

3.5 Fine-Tuning

SOTA IRC approaches have achieved strong performance by supervised fine-tuning of LLMs [13, 3, 2], using Low-Rank Adaptation (LoRA) [14]. In these studies, the model is trained on labeled issue reports and predicts the labels for unseen test issues. Each training example is formatted as an instruction prompt which includes the issue's title and description as input, and its label as output. At inference, the trained model receives the test issue in the same format, and predicts the empty label field. Regular expressions are used to extract the prediction from the output, and unparseable outputs are marked as *invalid*. We use this established methodology as our fine-tuning baseline.

For memory efficiency, each model is loaded in 4-bit quantization with Unsloth[2]. The LoRA rank and alpha are set to 16, with a learning rate of 2×10^{-4} and the paged AdamW 8-bit optimizer. Training runs for 3 epochs with a per-device batch size of 1 and gradient accumulation steps of 16. The max sequence length is set to 2,048 tokens, which covers 94.9% of the issues in our dataset. Issues that exceed this budget get truncated from the right side of their description text to fit.

4 Experimental Setup

4.1 Dataset and Settings

We use the dataset introduced in Heo et al. [13] for our experiments. The dataset has a total of 6,600 issue reports gathered from eleven active open-source projects: `ansible/ansible`, `bitcoin/bitcoin`, `dart-lang/sdk`, `dotnet/roslyn`, `facebook/react`, `flutter/flutter`, `kubernetes/kubernetes`, `microsoft/TypeScript`, `microsoft/vscode`, `opencv/opencv`, and `tensorflow/tensorflow`. Each project has 600 issue reports with an equal distribution across the labels (*bug, feature, question*). To support their LLM fine-tuning method, Heo et al. divided the dataset into 50/50 train and test splits, stratified by project and label. This results in 300 train and 300 test issue reports per project.

We also adopt Heo et al.'s project-specific (PS) and project-agnostic (PA) configurations to examine how data scope affects IRC performance. In the PS setting, each project's train

[1] Squared-similarity weighting and uniform majority were also evaluated, but were outperformed by similarity-weighted voting.

[2] `https://unsloth.ai`

and test data are used in isolation, resulting in eleven different train-test pairs. In the PA setting, the train splits of all projects are merged into a single training set of 3,300 issue reports. In both configurations, RAGTAG and VOTAG only use the training splits as their retrieval index to ensure fair comparison with the fine-tuning baseline.

4.2 Models

We use the Qwen2.5-Instruct model family [27] in our evaluations with four different parameter sizes: 3B, 7B, 14B, and 32B. Qwen2.5 is an open-source LLM with strong performance on general-purpose benchmarks. Using the same model at different parameter sizes isolates the effect of model scale from model architecture across LLM-based IRC approaches. Model temperature is set at 0.1 for deterministic classifications.

4.3 Evaluation Metrics

We report macro-averaged precision, recall, and F_1 over the three labels (`bug`, `feature`, `question`), along with overall accuracy. Per-class metrics are reported when the asymmetry across classes is itself the subject of analysis. Outputs that fail to parse a valid label (after the XML-tag extraction and regex fallback described in **??**) are counted as incorrect, and an *invalid rate* is reported separately so any parsing-induced loss is auditable.

4.3.0.1 Aggregation convention (PA vs PS).

The PA setting produces a single set of predictions over the full 3,300-issue test set, and we compute macro F_1 directly over that set. The PS setting produces eleven sets of predictions, one per project, each over its own ∼300-issue test slice. Throughout the paper we report PS results using *pooled* aggregation: the eleven per-project prediction sets are concatenated into one 3,300-issue test set and macro F_1 is computed once on the pool. This treats each test issue as one observation in both settings and yields apples-to-apples comparisons between PA and PS. The alternative — computing macro F_1 separately within each project and averaging the eleven numbers — weights projects rather than issues; because macro F_1 is non-linear in the underlying confusion-matrix counts, the two conventions disagree, and on our data the per-project mean is consistently 0.002–0.011 macro F_1 below the pooled value with the gap largest for fine-tuning. We adopt pooling as the canonical reporting convention to remove this asymmetry; per-project breakdowns appear only where the project is the unit of analysis and are explicitly labelled as such.

4.4 Setup Hardware

5 Evaluations

5.1 RQ1) To what extent can similarity-weighted voting on the nearest neighbors (VOTAG) classify issue reports, and at what point does adding neighbors stop helping?

We evaluate VOTAG's performance on both PA and PS settings at k values 1 to 30. The resulting macro F1 scores are shown in Figure 1. VOTAG's best macro F1 is 0.595 (k=15) in PS and 0.604 (k=16) in PA. Both settings reach 99.8% (PS) and 98.5% (PA) of their respective peak by k=7. When adding neighbors beyond the peak, PS loses 0.015 and PA loses 0.011 by k=30.

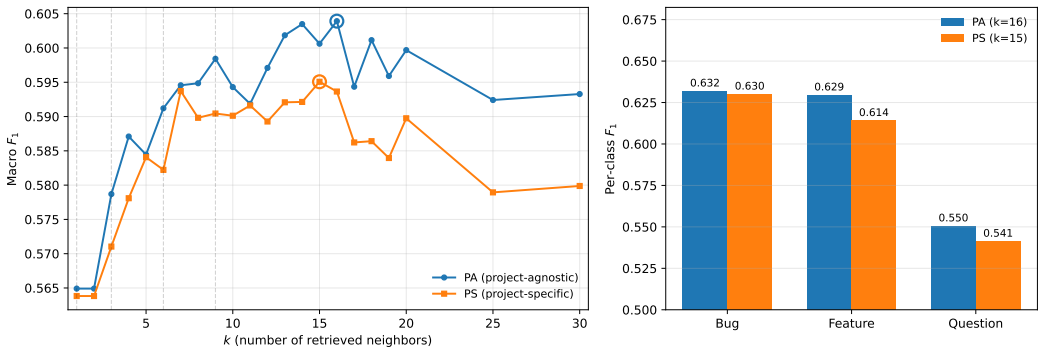

Figure 1 (left) VOTAG macro F_1 as a function of k in both PS and PA settings. Circles mark peak performance. **(right)** Per-class F_1 at each setting's best k (k=15 for PS, k=16 for PA).

PA consistently outperforms PS at every k. However, the performance gap is marginal: averaging 0.007 macro F1, with a maximum of 0.015 at k=18. This similar performance is due to the large overlap of the top-k neighbors in both settings. Specifically, PA and PS share 84–90% of the top-k neighbors across $k \in [3, 30]$. Therefore, even when using the full 3,300-issue retrieval index, most of the retrieved neighbors are from the same project. This means issue reports from the same project tend to be closer in the embedding space.

Question consistently has the weakest performance among the three labels at every k in both settings. At best k, PS per-class F1 is 0.630 for bug, 0.614 for feature, and 0.541 for question. The corresponding PA values are 0.632, 0.629, and 0.550 (Figure 1). Additionally, questions are more frequently misclassified as bugs: 32% of question issues are predicted as bugs in both settings, while 18% (PS) and 17% (PA) are predicted as features. Since VOTAG predicts the label with the most accumulated similarity weight, this misclassification asymmetry indicates that question issues are more similar to bugs than to features in the embedding space.

5.2 RAGTAG

We evaluate RAGTAG at k values $\{0, 1, 3, 6, 9, 12, 15\}$ on both PA and PS settings, where k=0 is the zero-shot baseline [7]. k value grid selection is informed by VOTAG's peak performance around k=15 in the previous section. Figure 2 shows the macro F1 of every model across this grid for both settings.

PS marginally outperforms PA at almost every (model, $k \geq 1$) pair evaluated (zero-shot is identical between settings since no retrieval is performed). The close performance gap is due to the same top-k neighbor overlap observed in the VOTAG analysis. The marginal improvement is significant because PS uses 11× less retrieval data per project: each PS index covers only its own 300 retrieval issues, while PA uses the full 3,300 across all 11 projects. Given this practical advantage, we conduct the rest of our RAGTAG analysis on the PS setting and report PS numbers throughout this section.

The best k grows with model size: Qwen-3B peaks at k=3 (macro F1 = 0.697), Qwen-7B at k=6 (0.718), and both Qwen-14B and Qwen-32B at k=12 (0.732 and 0.767 respectively). This shows that larger models can use the context from additional examples more effectively. However, adding neighbors past k=12 has no benefit and slightly degrades performance: every model loses 0.001–0.015 macro F1 going from k=12 to k=15.

Every RAGTAG configuration with $k \geq 1$ outperforms the zero-shot baseline. At each

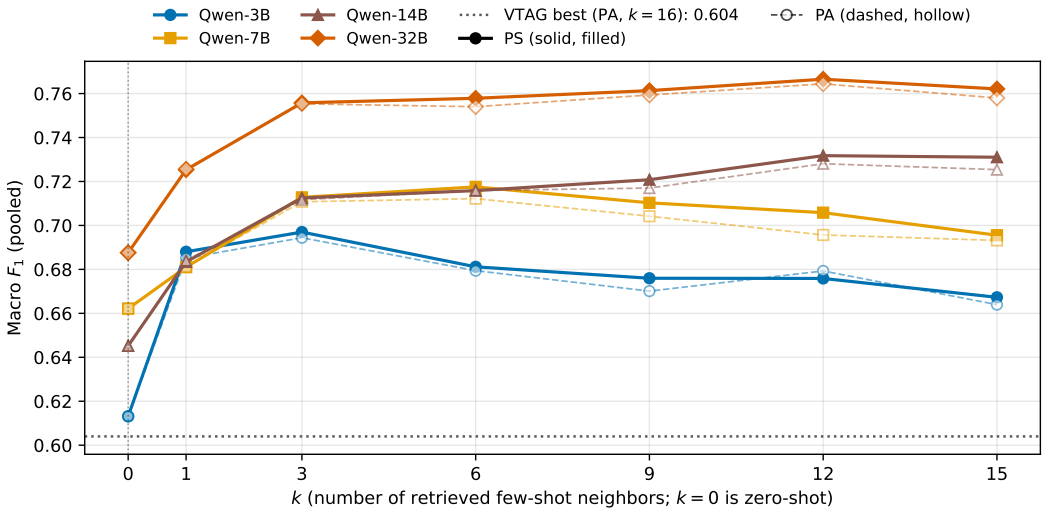

Figure 2 RAGTAG macro F_1 as a function of k for all four Qwen sizes. k=0 is the zero-shot baseline. The dotted horizontal line marks VOTAG's best (0.604, PA k=16).

model's peak, the gap over zero-shot averages +0.076 macro F_1 and ranges from +0.055 (Qwen-7B) to +0.087 (Qwen-14B). This shows that LLMs benefit substantially from the most similar labeled examples for IRC.

Every RAGTAG configuration outperforms VOTAG's best result (0.604 at PA k=16), with the smallest margin from Qwen-3B zero-shot (F1 = 0.613, +0.009) and the largest from Qwen-32B PS at k=12 (F1 = 0.767, +0.163). These results show that adding the LLM reasoning improves IRC on top of pure retrieval.

Similar to VOTAG, question consistently has the weakest performance among the three labels across all RAGTAG settings. Figure 3 shows per-class F_1 at each model's best k. Even the smallest gaps between question and the other labels, observed at Qwen-32B with k=12, are noticeable: question is 0.060 macro F_1 below bug and 0.143 below feature. Similar to the VOTAG observations, questions tend to be more frequently misclassified as bugs. At zero-shot, across all four model sizes, 46–59% of true-question issues are predicted as bugs while only 5–16% are predicted as features. RAGTAG narrows this misclassification asymmetry but does not eliminate it: at each model's best k, 26–36% of question issues are still predicted as bugs and only 9–15% as features.

These results show that the LLMs are already prone to misclassifying question issues as bugs, even before the top-k examples are used. Therefore, presenting bug-labeled examples for true-question query issues likely helps the misclassification. Based on this intuition, we propose a more balanced retrieval technique as an intervention to reduce question-to-bug misclassifications. To prevent damaging performance on true-bug issues, we apply this neighbor balancing only when the query is suspected to be a question. We use the label distribution of the retrieved top-k neighbors as the suspicion signal. Averaged across $k \in [1, 30]$, question query issues retrieve roughly balanced bug and question examples (mean $N_{\text{bug}} - N_{\text{question}} = -1.4$). Therefore, We treat a query issue as a suspected question when the question count in the top-k is within a margin m of the bug count. We select m=3 empirically

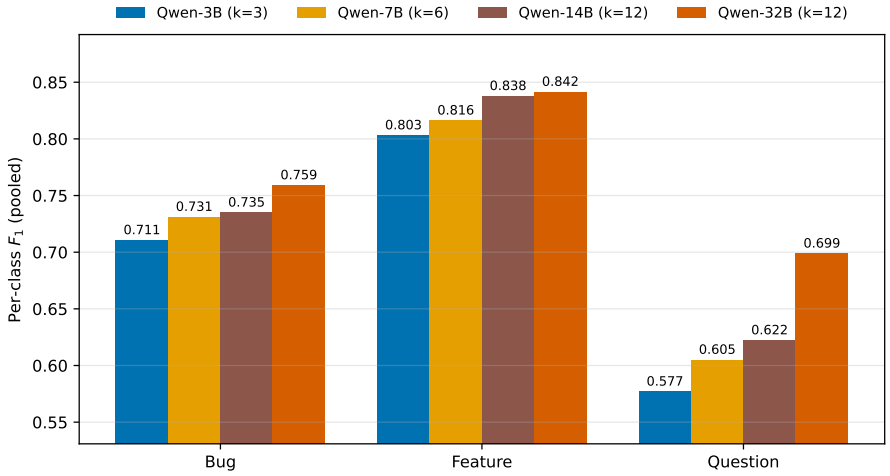

Figure 3 Per-class F_1 for each model size at its best RAGTAG-PS configuration.

from $m \in 1, 2, 3, 4, 5$[3]. Averaged across $k \in [1, 30]$, m=3 fires for 79% of true-question issues while firing for only 41% of true-bug issues. This trade-off is more favorable than other margin values. We further confirmed this choice with a margin sweep on Qwen-3B (the smallest model), where m=3 had the best performance.

6 Discussion

7 Threats to Validity

8 Conclusion

9 Data Availability

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